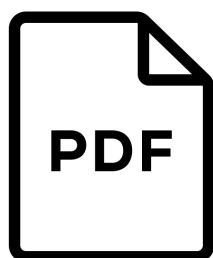


# CaSSius: a PDF typesetter using CSS regions (via polyfill)

**Cite as:** Martin Paul Eve, "CaSSius: a PDF typesetter using CSS regions (via polyfill)", *eve.gd*: Martin Paul Eve, July 23, 2015, <https://doi.org/10.59348/ny4q3-hpt21>.

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## CaSSius

**A**nnouncing [CaSSius](#): a tool to create beautiful paginated PDF documents from HTML content using CSS regions. It is intended to be part of [XML-first/XML-in workflows](#) for scholarly communications but may have alternative uses.

CaSSius: heavyweight typesetting with lightweight technology.

#Usage and Quick Start Guide CaSSius takes standard HTML content in a pre-specified form and flows it between CSS regions (see “document structure” below). To begin using CaSSius follow these steps:

1. Move or copy the “cassius” directory from this repository into the root of your website.
2. Insert the following code into the head tag of your HTML document:

```
<link rel="stylesheet" href="cassius/cassius.css">
<link rel="stylesheet" href="cassius/cassius-content.css">
<script type="text/javascript" src="cassius/jquery.js"></script>
<script type="text/javascript" src="cassius/cassius.js"></script>
<script src="cassius/regions/css-regions-polyfill.min.js"></script>
```

3. Optionally, insert the javascript code for [Adobe typekit](#) (you will need to sign up for an account).
4. Format your document according to the “document structure” guide below.
5. Replace cassius/images/logo.png with your own logo.
6. Load the page in a browser, wait until it has finished typesetting and then use the browser’s “print to PDF” option to create your document.

#Document Structure The following rules should be strictly adhered to in order to produce correct documents.

Every HTML file should contain a CaSSius metadata block. A CaSSius metadata block should be wrapped inside a script tag with type set to “text/cassius” and an id attribute of “cassius-metadata. A CaSSius metadata block may contain the following elements:

```
<script type="text/cassius" id="cassius-metadata">
  <div id="cassius-metadata-block">
    <div id="cassius-title">Article typeset by CaSSius: heavyweight typesetting
with lightweight technology</div>
    <div id="cassius-publication">CaSSius</div>
    <div id="cassius-authors">Martin Paul Eve</div>
    <div id="cassius-affiliations">Department of English and Humanities, School
of Arts, Birkbeck, University of London, United Kingdom</div>
    <div id="cassius-doi">10.16995/olh.001</div>
    <div id="cassius-date">September 2015</div>
  </div>
</script>
```

If the “cassius-title” metadata div is not present, CaSSius will use the HTML document’s “title” element in the “head” of the document.

The basic structure of a CaSSius document is as follows (also available in [template.html](#)):

```

<body>
  <div id="cassius-content">
    <h1 class="articletitle"></h1>
    <div class="authors"></div>
    <div class="affiliations"></div>

    <div class="abstract">
      <h2>Abstract</h2>
      <p>Your abstract content here.</p>
      <p>As many paragraphs as needed.</p>
      <p class="oa-info">&copy; 2015 Martin Paul Eve. This is an open-access
article distributed under the terms of the Creative Commons Attribution License, which
permits unrestricted use, distribution, and reproduction in any medium, provided the
original author and source are credited.</p>
    </div>

    <div class="main">
      <div class="section">
        <h1>A section title</h1>
        <p>A paragraph.</p>
      </div>

      <div class="section">
        <h1>A section title</h1>
        <p>A paragraph with a footnote.<a href="#fn1--fragment" id="xr1">
<sup>1</sup></a></p>
      </div>

      <div class="notes">
        <h1>Notes</h1>
        <div class="footnote"><p><span class="generated"><a href="#xr1--fragment"
id="fn1">1</a></span> Footnote content goes here.</p></div>
      </div>

      <div class="references">
        <h1 class="ref-title">References</h1>
        <div class="section ref-list">
          <ul>
            <li class="ref-content">Adorno, Theodor W., <i>Negative
Dialectics</i>, trans. by E.B. Ashton (London: Routledge, 1973)</li>
          </ul>
        </div>
      </div>
    </div>
  </div>
</div>

```

```
<article id="article"></article>

<script type="text/cassius" id="cassius-metadata">
  <div id="cassius-metadata-block">
    <div id="cassius-title">Article typeset by CaSSius: heavyweight typesetting
with lightweight technology</div>
    <div id="cassius-publication">CaSSius</div>
    <div id="cassius-authors">Martin Paul Eve</div>
    <div id="cassius-affiliations">Department of English and Humanities, School
of Arts, Birkbeck, University of London, United Kingdom</div>
    <div id="cassius-doi">10.16995/olh.001</div>
    <div id="cassius-date">September 2015</div>
  </div>
</script>
</body>
```

#Import from JATS/NLM An early-stage version of an import function from JATS is implemented in [cassius-import/bin/cassius-import.py](#). This script requires python and java.

Usage:

```
cassius-import.py <in-file> <out-file> [options]
cassius-import.py (-h | --help)
cassius-import.py --version
```

A sample XML file to show this working (and the scope of implementation to date) [can be found in the cassius-import directory](#).

#Performance and Settings If you are consistently typesetting documents that are over fifty pages long, you may see a performance increase if you change the value of initialPages to a higher setting in cassius.js. Setting this to a higher value will yield better performance on larger documents, but worse performance on smaller documents.

#Components and Licensing CaSSius is copyright Martin Paul Eve 2015. It is released under the terms specified in [LICENSE](#).

CaSSius makes use of several other open-source/free-software projects, including:

- [css-regions-polyfill](#). Copyright (c) 2014 François REMY with [a BSD-style license](#).
- [jQuery](#). Under [the MIT license](#).
- The CaSSius logo is a derivative of a work by [Lil Squid from the Noun Project](#), licensed under the Creative Commons Attribution License.
- Parts of the cassius-import library contain materials from the National Library of Medicine, specifically [adaptations of their XSLT suite and entity resolution files, which are public domain](#).
- The cassius-import library links to the Saxon XSLT and XQuery Processor from Saxonica Limited <http://www.saxonica.com/> which is dual-licensed under the GNU GPL and the Mozilla Public License <http://www.mozilla.org/MPL/>. This dual-licensing applies ONLY to the file saxon9.jar.